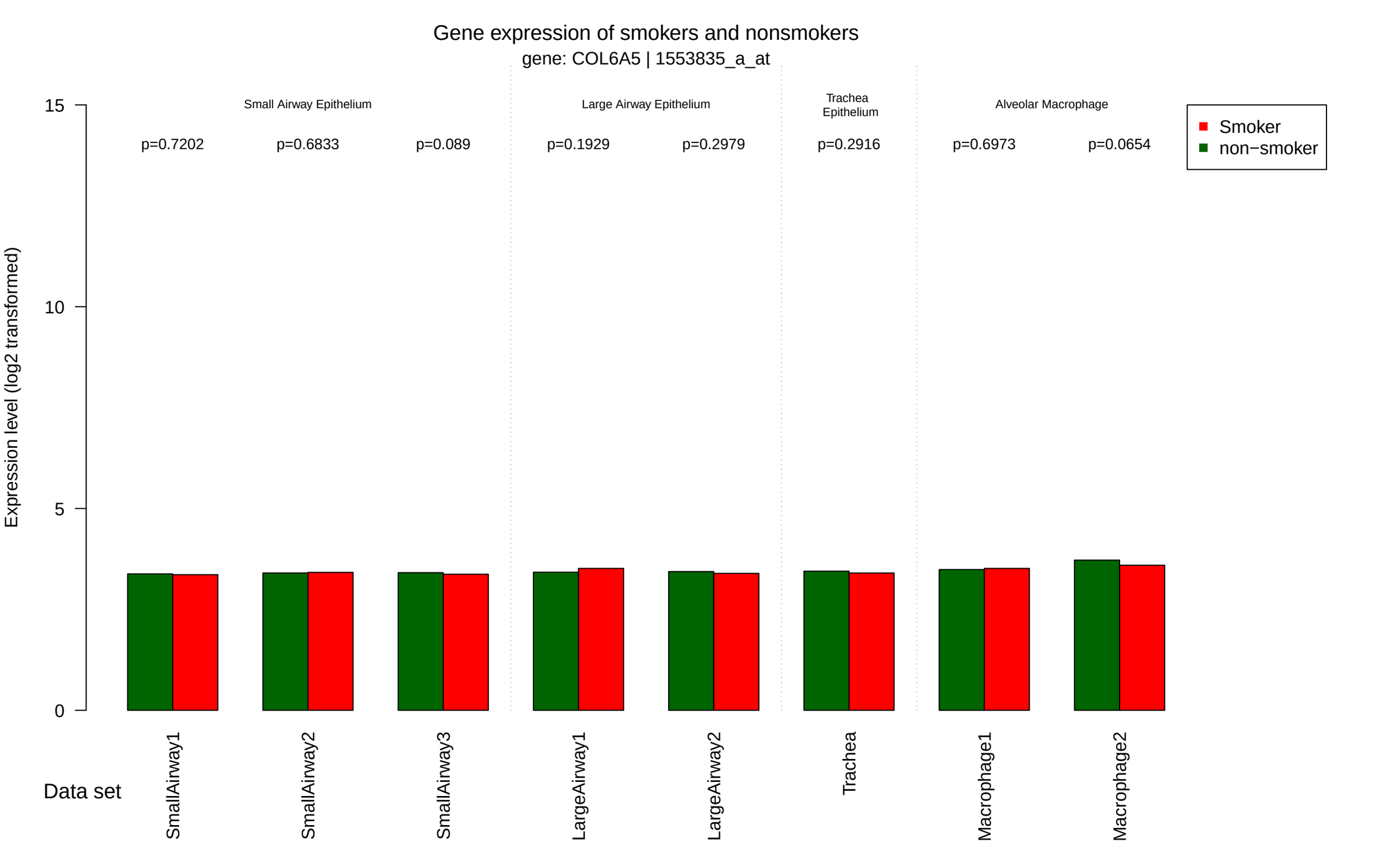
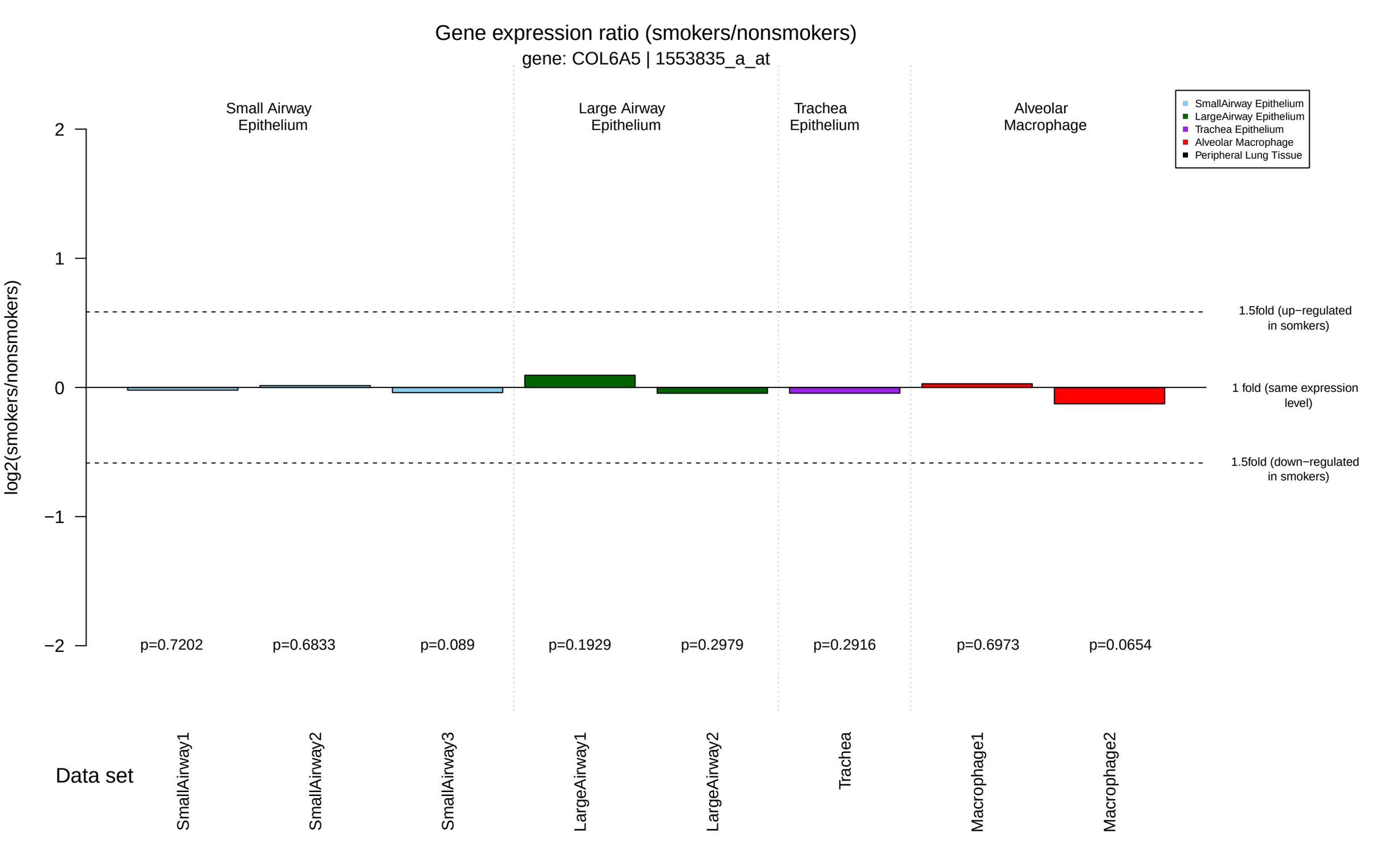
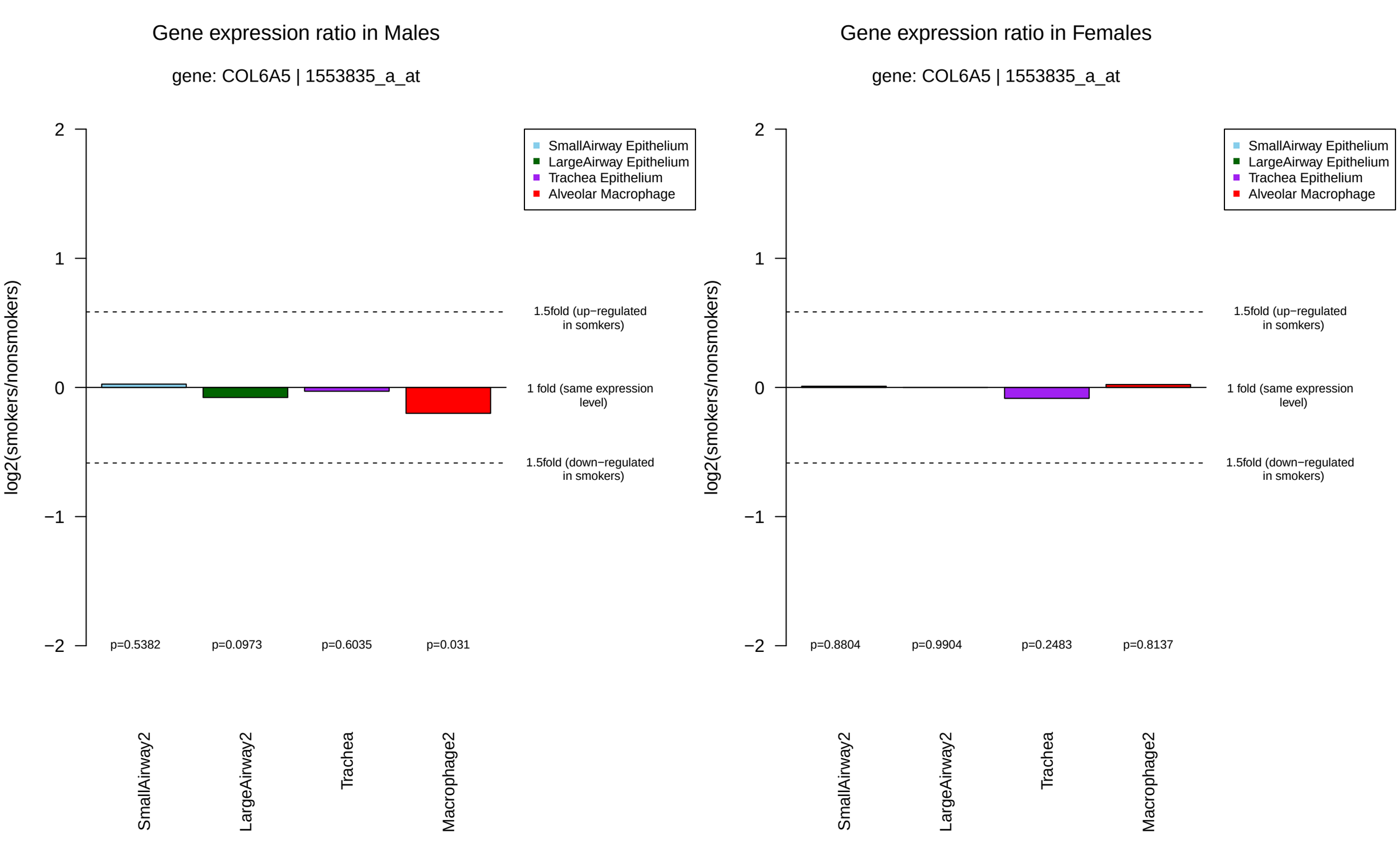
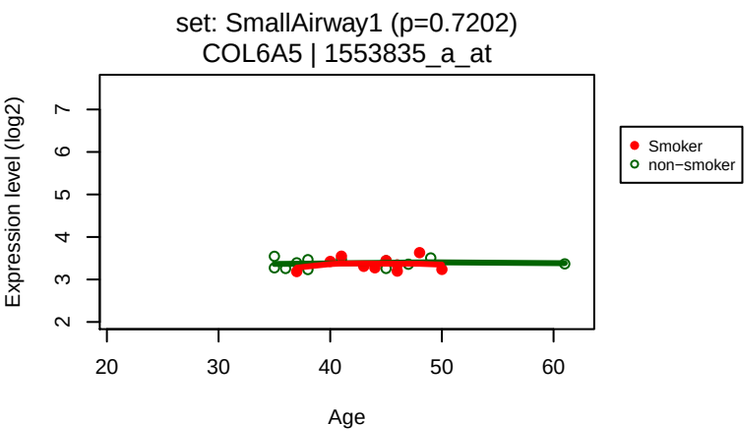
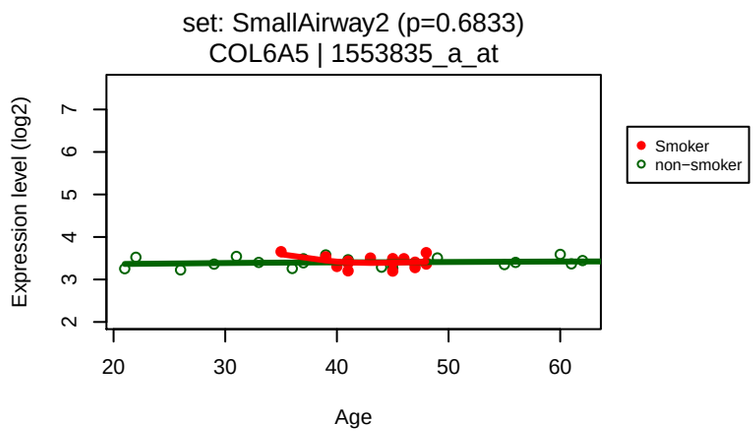
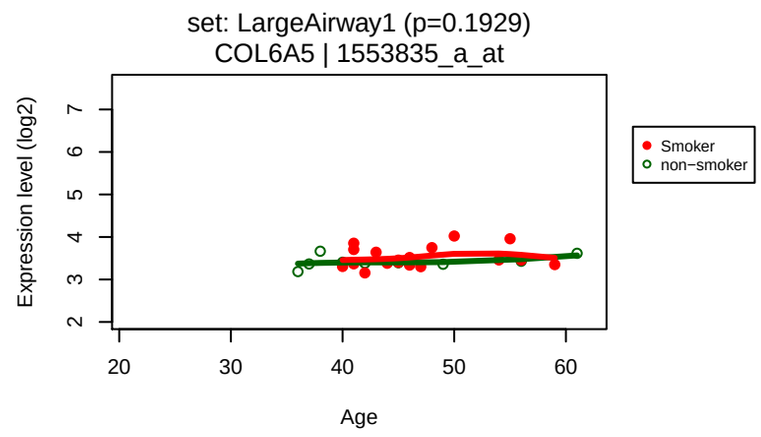

Gene expression of smokers and nonsmokers
gene: COL6A5 | 1553835_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7202
p=0.6833
p=0.089
p=0.1929
p=0.2979
p=0.2916
p=0.6973
p=0.0654
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

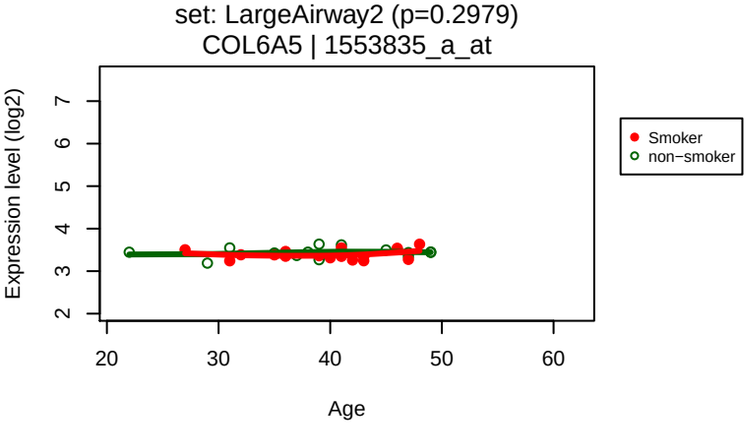

Gene expression ratio (smokers/nonsmokers)
gene: COL6A5 | 1553835_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7202
p=0.6833
p=0.089
p=0.1929
p=0.2979
p=0.2916
p=0.6973
p=0.0654
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

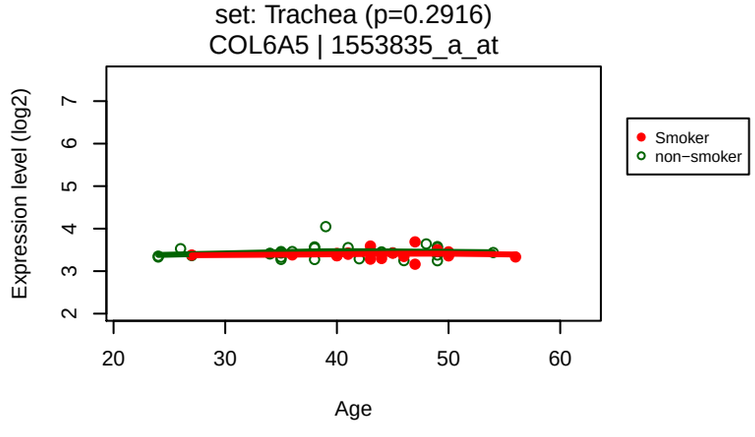

Gene expression ratio in Males
gene: COL6A5 | 1553835_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5382
p=0.0973
p=0.6035
p=0.031
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL6A5 | 1553835_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8804
p=0.9904
p=0.2483
p=0.8137
SmallAirway2
LargeAirway2
Trachea
Macrophage2

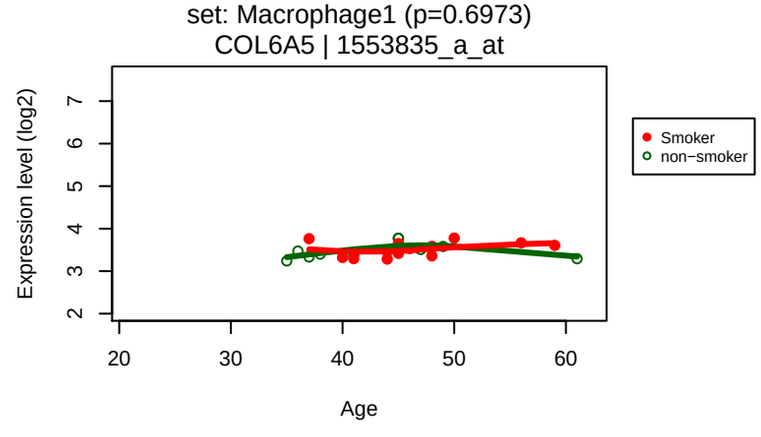

set: SmallAirway1 (p=0.7202)
COL6A5 | 1553835_a_at
Expression level (log2)
Age
Smoker
non−smoker

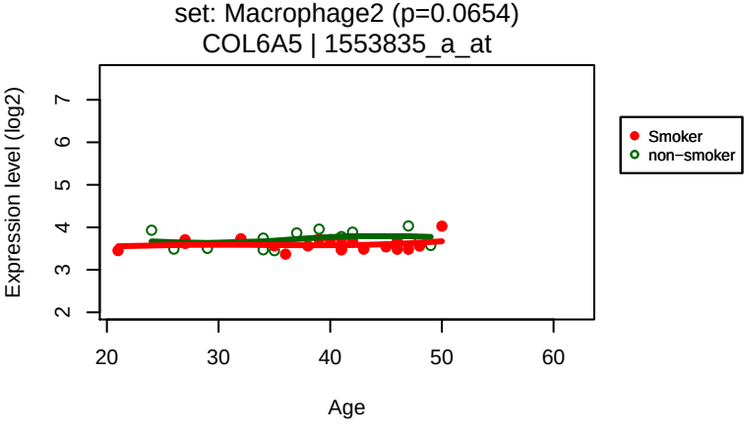

set: SmallAirway2 (p=0.6833)
COL6A5 | 1553835_a_at
Expression level (log2)
Age
Smoker
non−smoker

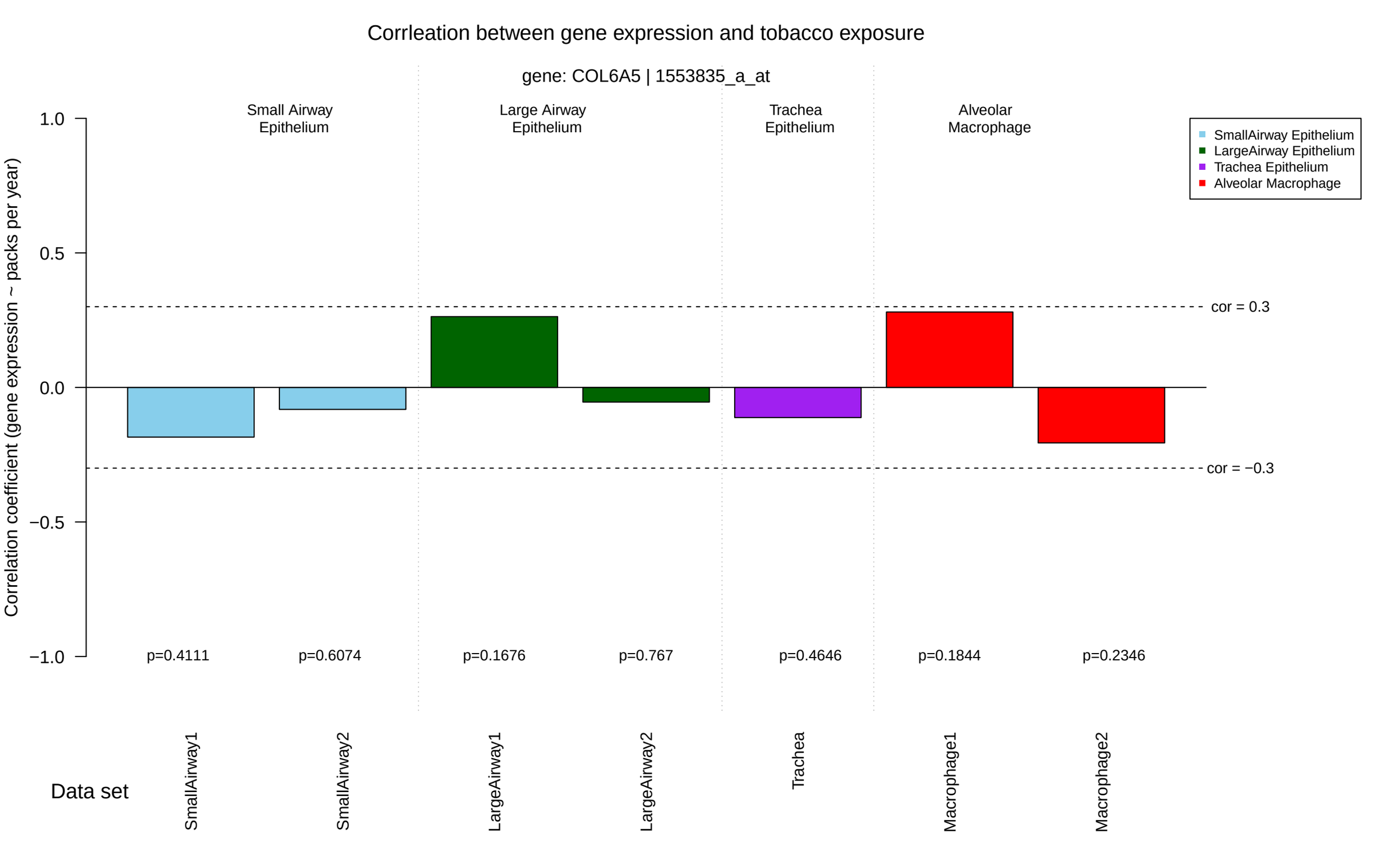

set: LargeAirway1 (p=0.1929)
COL6A5 | 1553835_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2979)
COL6A5 | 1553835_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2916)
COL6A5 | 1553835_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6973)
COL6A5 | 1553835_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0654)
COL6A5 | 1553835_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL6A5 | 1553835_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4111
p=0.6074
p=0.1676
p=0.767
p=0.4646
p=0.1844
p=0.2346
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2